

WASP-36b

R-0763

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_250118 · created 2026-07-28T04:17:28+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460693.8871 +/- 0.006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.047
Transit depth (Rp/Rs)^2	2.78 +/- 1.58 [%]
Orbital Inclination (inc)	85.06 +/- 2.93
Airmass coefficient 1 (a1)	0.172 +/- 0.098
Airmass coefficient 2 (a2)	1.14 +/- 0.6
Scatter in the residuals of the lightcurve fit is	53.64 %
Best Comparison Star	#2 - [565, 66]
Optimal Aperture	2.41
Optimal Annulus	3.0
Transit Duration (day)	0.084 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.167 $\pm$ 0.047 vs 0.1368 (deviation 0.49 σ)
Duration fit vs published	0.084 d vs 0.0758 d (deviation 0.34 σ)
Mid-time O-C	13.9 min
Depth SNR	2.7
Residual scatter	53.64 %

Light curve

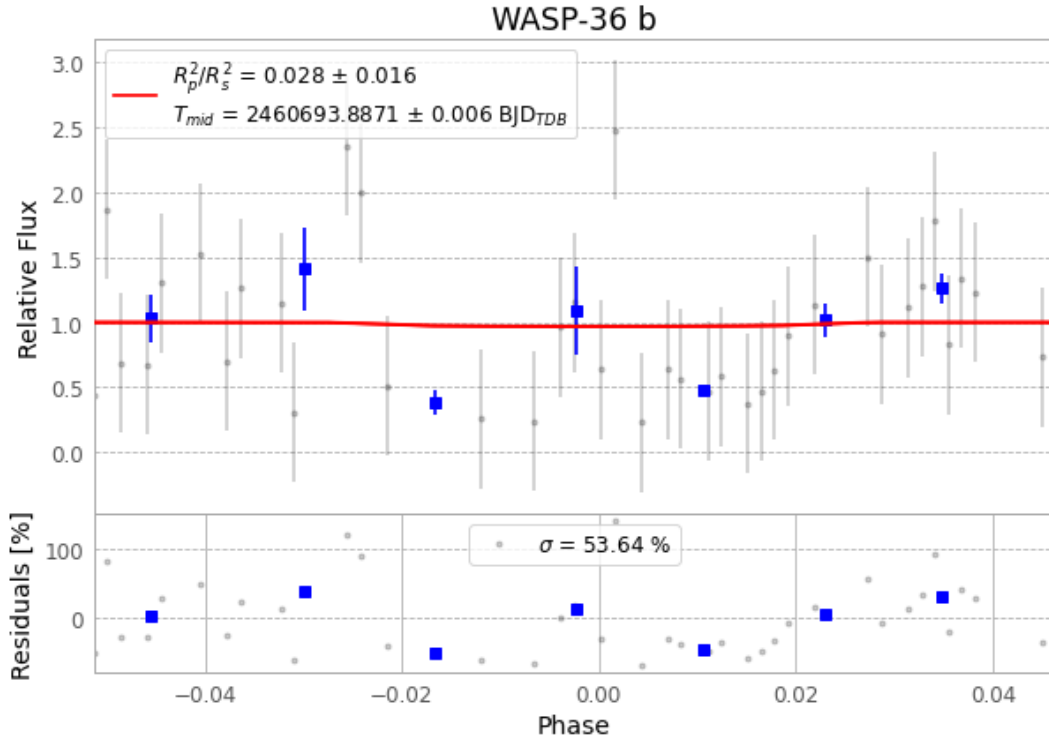


Figure 1 — FinalLightCurve_WASP-36 b_2025-01-18.png (SHA-256 dcd6b9e6c08f148a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [534, 409], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d3a7dba2082e33e2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

73 FITS frames (48 MB), WASP-36250118071516.FITS → WASP-36250118105116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-250118.log (114d19402886...), FinalLightCurve_WASP-36 b_2025-01-18.png (dcd6b9e6c08f...), FinalLightCurve_WASP-36 b_2025-01-18.csv (3c543fa04199...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 04:17Z.